NAU NORTHERN ARIZONA UNIVERSITY
Steve Sanghi College of Engineering

Memorandum

MA5 - Sensors & Actuators

to: course staff

from: project team

context: undergraduate engineering project

Re: MA5 - Sensors & Actuators

1.1) What is the impact of turning on the motor carrier switch from off to on?

When the motor carrier switch is turned on, there is a green light on the right side of the board, and the board can operate without being plugged in to power. When the switch is off, the board needs to have its power supplied from the USB.

1.2) Which of the 4 motor objects (M1 to M4) corresponds to the flywheel motor? Look at the motor carrier board. Does that object correspond with the labeling on the board?

Motor object M3 corresponds to the flywheel motor. This makes sense, as the flywheel motor's red and black cables connect to terminals labeled M3+ and M3-.

1.3) What is the connection between the LED brightness and motor speed?

The LEDs got brighter as the duty cycle got further from zero. The sign of the duty cycle didn't matter, just the magnitude of the duty cycle.

2.1) What encoder object reports the flywheel wheel velocity? Look at the motor carrier board. Does that object correspond with the labeling on the board?

Encoder1 reports the flywheel velocity. This corresponds with the labeling on the board, because the flywheel has cables connected to terminals labeled HA1 and HB1.

2.2) What is the resolution of the flywheel encoder and the geared motor encoder?

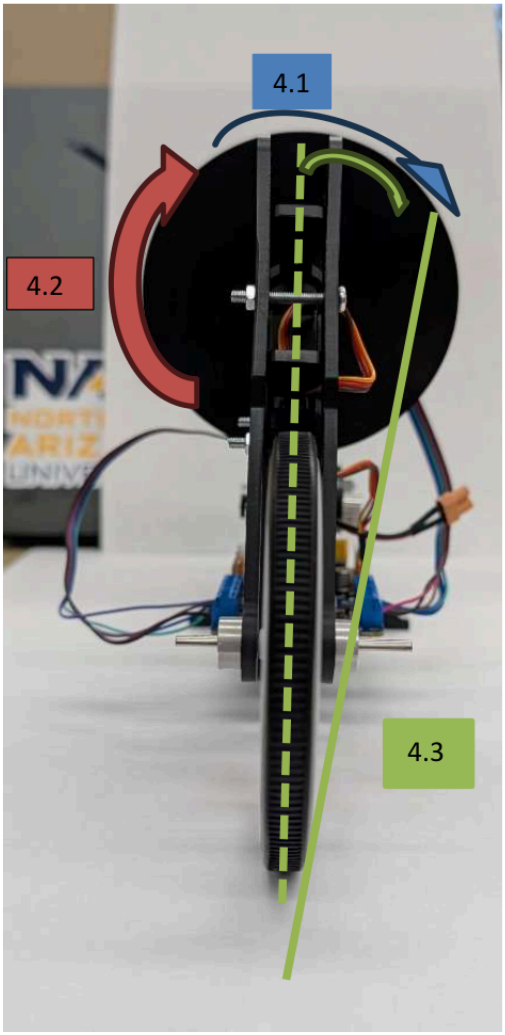
The flywheel encoder has a resolution of ~48 points per rotation, and the geared motor encoder has about 1200 points of resolutionn

3.1) Which Euler angle corresponds to tilt of the motorcycle, and which is positive?

Roll corresponded to the tilt of the motorcycle. Tilting towards the motorcycle's left was positive.

3.2) Which gyro angle corresponds to the tilt velocity of the motorcycle, and which direction is positive?

Gyro y corresponds to the motorcycle's tilt velocity. Tilt velocity is positive toward the motorcycle's left.



4.1) Please annotate the photo of the front view of your motorcycle to indicate the direction of positive motor duty cycle.

4.2) Please annotate the photo of the front view of your motorcycle to indicate the direction of positive change in the encoder velocity (counts per second)

4.3) Please annotate the photo of the front view of your motorcycle to indicate the direction of positive tilt and tilt velocity measured by the IMU

5.1) Did your team negate appropriate sensor readings so that the direction of "positive" aligns for the 3 sensors and the flywheel motor?

Yes. In the serial monitor, all signs agree with eachother, and with the diagram to the left.